Source: https://usabilitygeek.com/paper-prototyping-as-a-usability-testing-technique/

CSE472: HCI – Human Computer Interaction

Lecture – 10: Prototyping

Spring 2026

BRAC UNIVERSITY

Inspiring Excellence

Paper Prototyping

- An essential tool in your design toolbox

How do we design things that actually fit user needs?

Problem:

- We can't evaluate a design until it's built

    But…

- After building, changes to the design are difficult
- What to do?

Solution

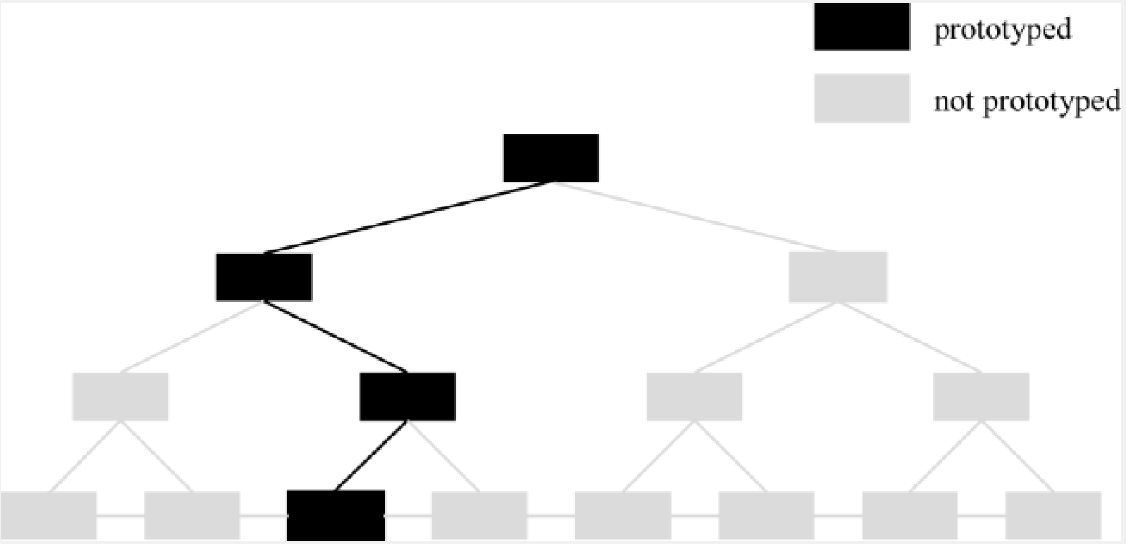
- Prototype!

Prototyping

Simulate the design in low-cost manner

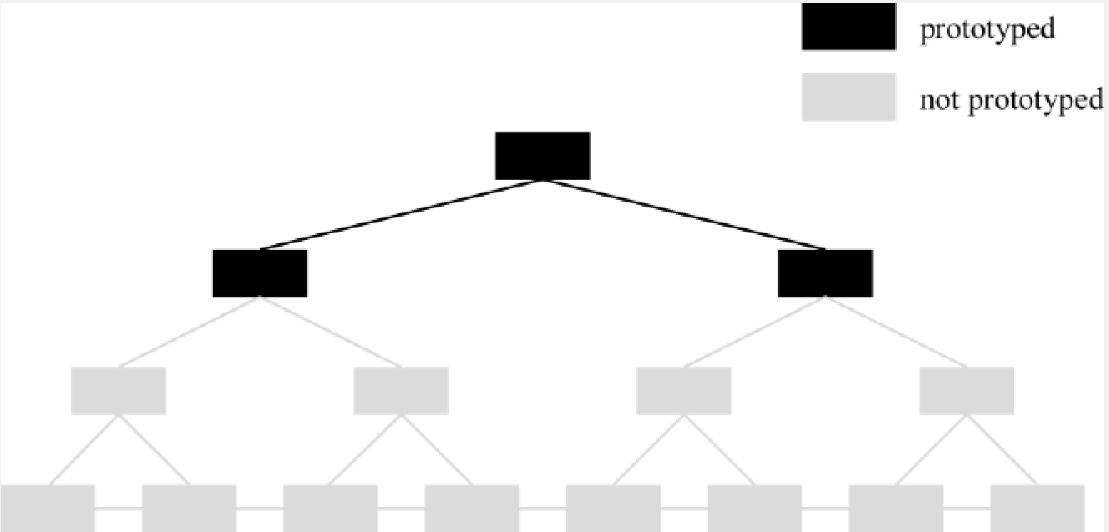
- Make it fast. Make it cheap.
- Facilitate iterative design and evaluation
  - Your first idea is rarely your best!
- Promote feedback
- Allow lots of flexibility for radically different designs
  - Don't kill crazy ideas!

How to prototype?



Vertical - "Deep" prototyping

- Show only portion of interface, but large amount of those portions

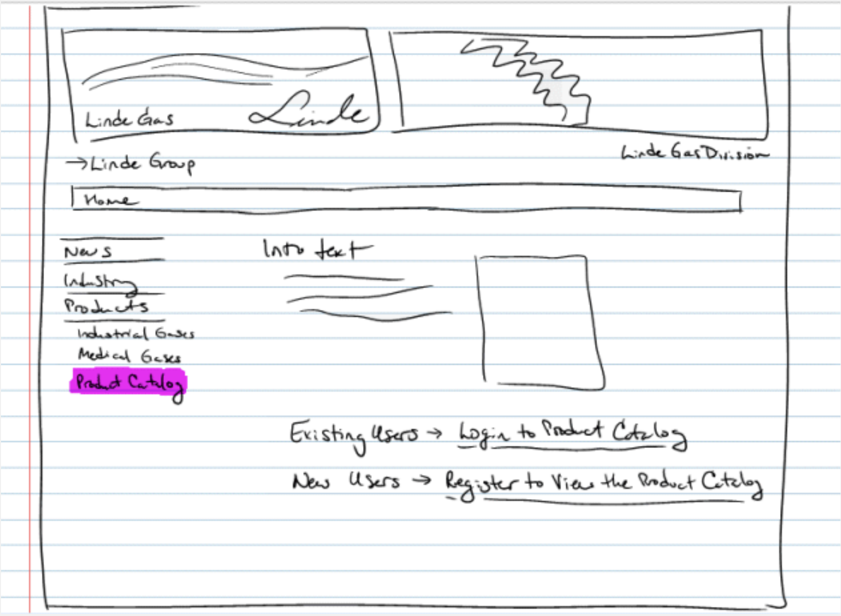


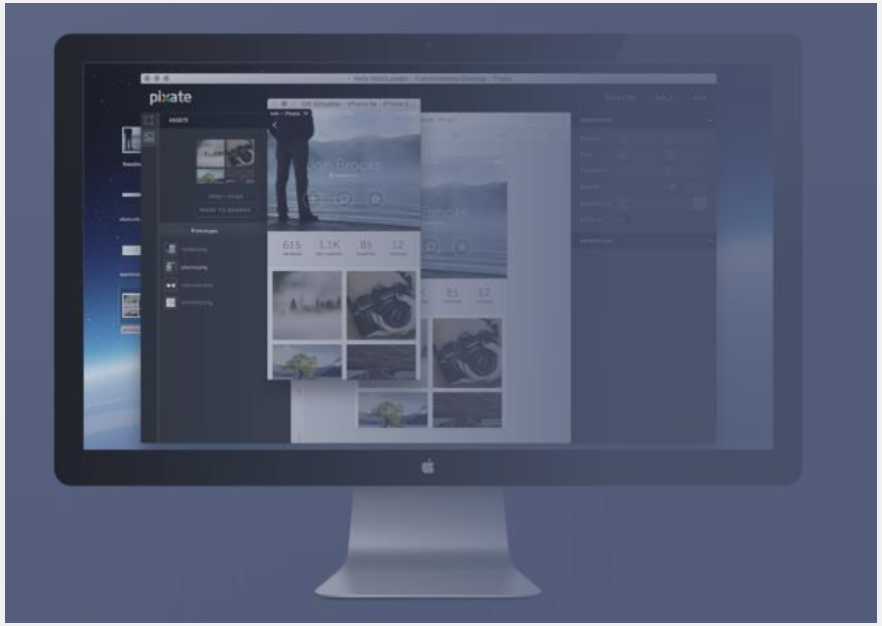
Horizontal - "Broad" prototyping

- Show much of the interface, but in a shallow manner

How to prototype?

Low fidelity vs. High fidelity





Amount of polish should reflect maturity of the prototype... Why?

- Low fidelity = explore ideas
- High fidelity = refine and validate details

“Mixed” fidelity

- Easy access to cameras makes it easy to blur the lines between lo-fi and hi-fi prototypes
- Photos of hand-drawn prototypes can easily be captured and displayed on real screens
- Sequences of photos can also be animated to simulate interaction

Today: Paper-Prototyping

An iterative design method where potential users perform realistic tasks by interacting with a paper version of the interface that is manipulated by a person 'playing computer,' who doesn't explain how the interface is intended to work.

Paper~Prototyping examples Video links

1. Mobile Application Design : Paper Prototype Video

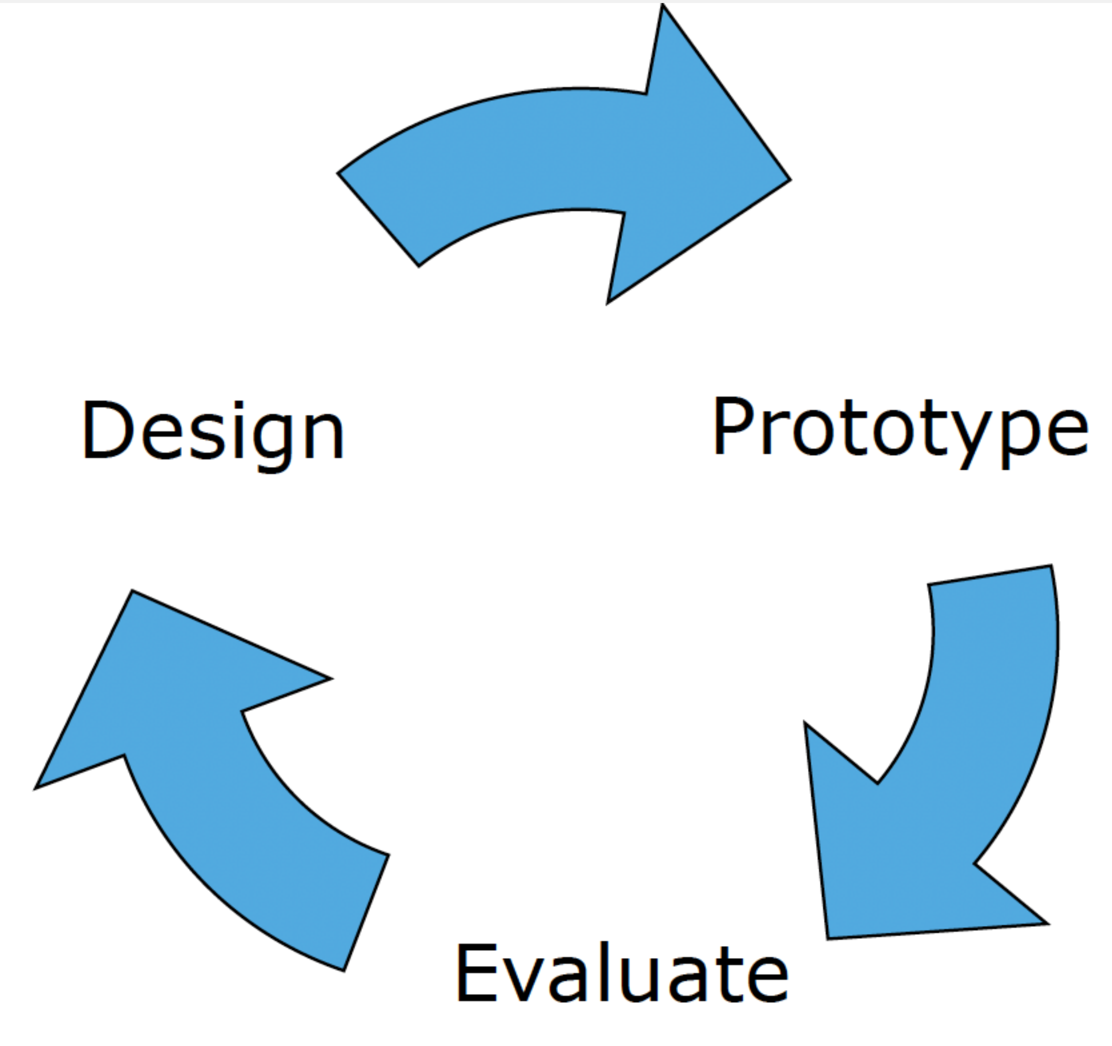
https://www.youtube.com/watch?v=y20E3qBmHpg

2. Rapid Prototyping: Sketching | Google for Startups

https://www.youtube.com/watch?v=JMjozqJS44M

Why do it?

- Principle of iterative design
  - Quality is partially a function of the number of iterations and refinements that a design undergoes



Why do it?

- Low cost
- Fast to implement
  - Typical hi-fi prototype takes a few weeks as opposed to a paper prototype that takes a few hours
- Allows you to merge the design and prototyping phase together
- It gets everyone involved!
  - Builds teamwork in groups with diverse skill sets
  - So simple, no one gets left out

Why do it?

Feedback on the BIG things

- *Lo-fi nature forces users to consider usability issues related to layout and control*
- *Nit picking over choice of colors, buttons sizes, font choice ignored*
- *Focus on Content as opposed to Appearance*

Why not to do it?

- *May seem unprofessional to some users*

  Maybe not the right prototype for the VCs ☺

- *Can't represent some effects with paper*
- *Typically, you would start with several rounds of paper prototyping, and move towards high-fidelity prototyping as the design becomes more finalized. i.e., YOU DO BOTH*

Building a Low-Fidelity Prototype

Gather essential materials

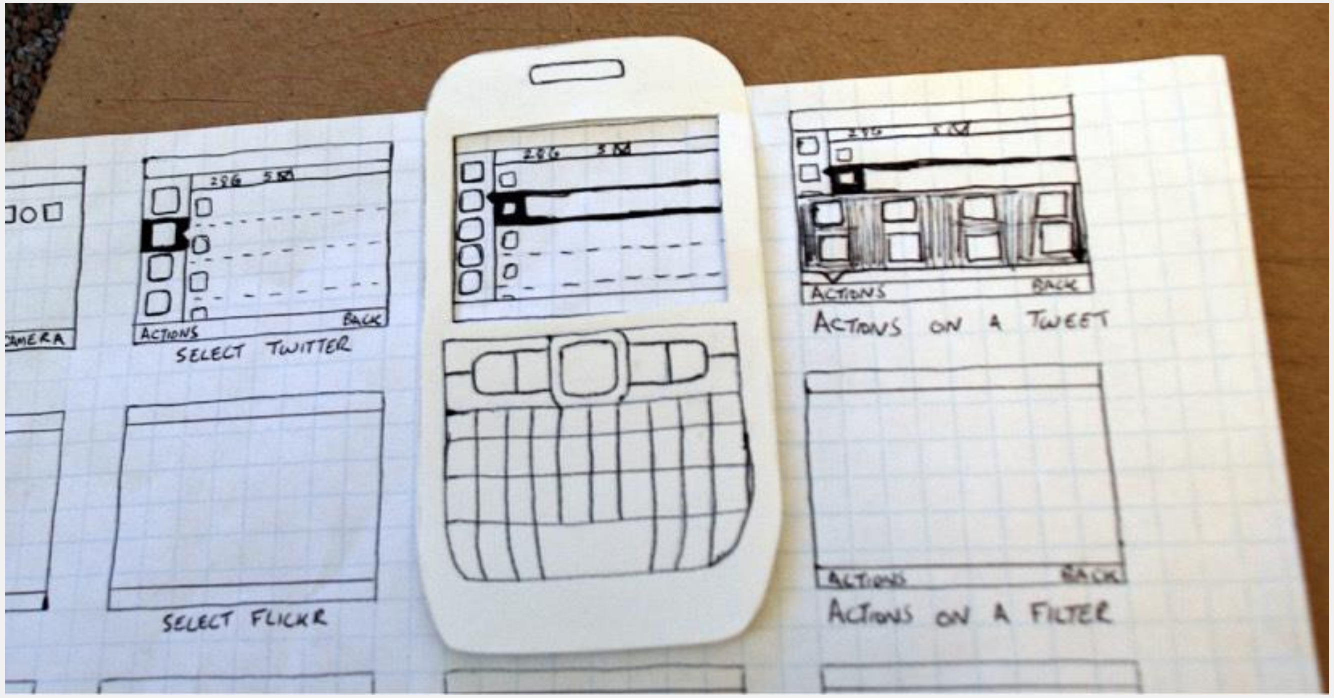
- White paper
- 5 by 8 inch cards
- Adhesives
- Markers
- Sticky pads
- Scissors
- Anything else you think of!!!

Building a Low-Fidelity Prototype

Don't get carried away with design!

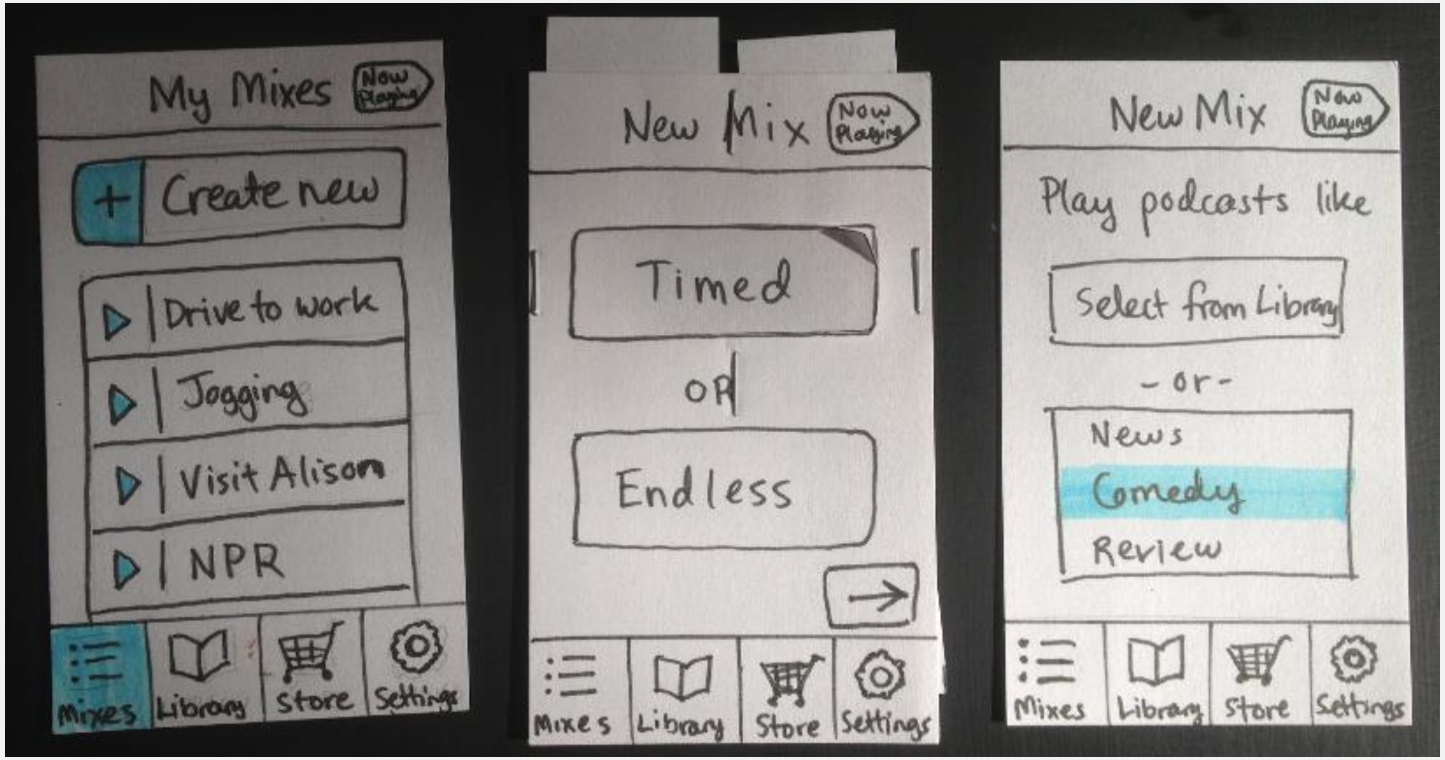
◦ Goal is to get as much user feedback as possible

◦ Set a deadline – forget minor details



Building a Low-Fidelity Prototype

- Draw generic frames
- Make everything needed to simulate effects
- Photocopier/camera is your friend!



Preparing for a test

Select users

- Perform user and task analysis
- Find out educational background, knowledge of computers, typical tasks required
- Get testers who fit the final user profile

Preparing for a test

Prepare test scenarios

Practice

- Sort out bugs/hitches before the real testing
- Get everyone comfortable with their role

Conducting a test

Facilitator

- Encourage user to express thoughts
- (don’t influence decisions!!!!)
- Giving instructions
- Making sure timing is met

Conducting a test

"Computer" person

- Arranges the paper prototype according to user input
- Needs to be organized
- Knows the prototype well
- Make changes quickly

Conducting a test

Observers

- Take notes
- Write possible solutions to problems faced
- Cannot react to user's actions

Evaluating Results

- *Summarize problems (e.g., make a list)*
  - *Usability issues*
  - *Missing (or mis-specified) functional requirements*
  - *Preferences for different alternatives*
  - *User priorities*
  - *Issues outside the user interfaces (e.g., high-level understanding)*
- *Prioritize problems*
- *Construct revised prototype*
- *Iterate, iterate, iterate!*

Summary: Paper Prototyping

An important prototyping tool (but not the only tool!)

- Quick to build/refine, thus enabling rapid design interactions. Useful tool for speeding up the process of iterative design
- Requires minimal resources and materials (cheap!)
- Detects usability problems at a very early stage before implementation.
- Focus on the “right” things early on
- Promotes communication between stakeholders. Team members gain understanding of user needs and priorities

I recommend you always do a few rounds of paper prototyping for every new design/app/system/solution that you create!

Any questions?

Acknowledgment: Lecture content adapted from Cornell University HCI & Design (Prof. Nicki Dell)